End to End Communication Transport layer

Exp No : 12
DATE : 10/10/26

Aim :
to Implement TCP & UDP echo (client & server) communication network programs using Socket programming.

-- server --
import socket
import threading
def start_server():
server = socket.socket(socket.AF_INET,
socket.SOCK_STREAM)
server.bind(('127.0.0.1', 5000))
server.listen(1)
print("[SERVER] listening on 127.0.0.1:5
conn, addr = server.accept()
print(f"[SERVER] connected by {addr}")
while True:
data = conn.v(1024).decode()
if not data or data.lower() == "quit":
print("[SERVER] Connection closed by client")
break
print(f"[SERVER] Received: {data}")
conn.send(f"Server received: {data}
encode

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    conn.close()
    server.close()

# --- client ---
import socket
import threading

def start_client():
    server = socket.socket(socket.AF_INET,
                           socket.SOCK_STREAM)
    server.bind
    server.listen()
    print("[SERVER] Listening ...")
    conn, addr = server.accept()
    print(f"[SERVER] Connected by {addr}")
    while True:
        msg = input("Enter message (or 'quit'): ")
        client.send(msg.encode())
        if msg.lower() == 'quit':
            break
        response = client.recv(1024).decode()
        print(f"[SERVER RESPONSE] {response}")
    client.close()

# --- main ---
if __name__ == "__main__":
    # Run server in a background thread
    server_thread = threading.Thread(target=
                                     start_server, daemon=True)
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server_thread.start()
# Run client in main thread
start_client()
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UDP

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import socket
import threading
# -- server --
def udp_server():
    s = socket.socket(socket.AF_INET, socket.
                      SOCK_DGRAM)
    s.bind(('127.0.0.1', 12000))
    print("[SERVER] Running on 127.0.0.1:12000")
    while True:
        data, addr = s.recvfrom(1024)
        msg = data.decode()
        if msg.lower() == 'quit':
            print(f"[SERVER] client {addr} ended the
                  connection")
            break
        print(f"[SERVER] Received from {addr}:
              {msg}")
    s.close()
    print("[SERVER] Stopped")

# -- client --
def udp_client():
    c = socket.socket(socket.AF_INET, socket.
                      SOCK_DGRAM)
    server = ('127.0.0.1', 12000)
```

```
while True:
    msg = input("Enter message (or 'quit' to stop): ")
    c.sendto(msg.encode(), server)
    if msg.lower() == 'quit':
        print("[Client] closing connection...")
        data = c.recvfrom(1024)
        print("Echo from server:", data.decode())
    c.close()
    print("[Client] stopped.")

# --- main ---
if __name__ == "__main__":
    threading.Thread(target=udp_server, daemon=True).start()
    udp_client()
```

Result:

The TCP and UDP client server program was implemented and executed successfully.



